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The epidemic triangle is comprised of the following three elements:

1. Agent
 - The microbe that causes the disease
2. Host
 - The organism affected
3. Environment
 - The external influences

The epidemic triangle explains that causation of infectious diseases is of those three major components which operate jointly.

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The subclinical stage of disease is an important factor to consider because the host is still contagious despite not having symptoms. This is important because it is important for the epidemiologists to consider that the host is still infected despite having no symptoms. For COVID, different health organizations took different approaches to the subclinical phase of COVID. In the US, this mainly took place as staying at home in isolation for the 2 week subclinical stage if you think you may have the disease (if someone you have frequent contact with ie a family member got COVID). For many other nations, they implemented a contact tracing system where if someone was infected then those close to them were mandated to also quarantine for 2 weeks to minimize the spread of infection.

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Isolation is for those who are known to have the disease. Quarantine is for those who may think they have the disease or may have been exposed to the disease. Isolation is a mitigative protective measure as it is mitigating the spread of the disease after the individual is known to have the disease. Quarantine is a preventative measure as it prevents the spread of disease before it is known for sure if an individual has the disease.

Specifically for COVID, the CDC recommends starting the isolation as soon as you test positive and to be super precautionary during the first five days of symptoms as that is when they are the most contagious. The precautions were in general just reduce the chances of others coming into contact with the one infected or their germs. The update I found for 2022 (with the

omicron variant) recommended that both quarantine and isolations to be a minimum of 5 days, with an additional 5 days of wearing a mask. However, for those who have moderate to severe systems, immunocompromised, or young children / unable to wear a mask should isolate / quarantine for longer periods.

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Infectivity of a disease refers to the agents ability to establish new infections in new hosts, meaning the disease is highly contagious with only a few particles of the disease. Pathogenicity refers to an agent's ability to make the host visibly sick, meaning the agent will make a large portion of the infected people to have visible symptoms.

These are two different measures as infectivity refers to an agents ability to make new infections while pathogenicity refers to an agents ability to make infected hosts have symptoms. Infectivity would affect the number of cases whereas pathogenicity affects how sick these cases are.

These are both important to consider for an infectious agent because agents with high infectivity will be best prevented by preventing or interrupting the transmission of the disease. High pathogenicity is more about creating strategies to protect the vulnerable.